

MSc Thesis Proposal

Open-Set Anomaly Detection of Crops and Weeds

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Motivation

Agricultural robots deployed for precision weeding are trained on a fixed set of known crop and weed species, but real fields inevitably contain plants outside that closed set (e.g., local weeds or species simply missing from the training data). A model that misclassifies these as one of its known classes risks either damaging a crop or missing a weed that should be removed. Open-set anomaly detection addresses this by teaching a model to recognize when a detection does not belong to any known class, rather than forcing a wrong label onto it. Unlike prior work on this topic that relied on a single target species against a relatively uniform background, the CropAndWeed dataset offers a genuinely alternative: 74 annotated crop and weed species across thousands of real field images, with a long-tailed species distribution that provides a natural split between common and rare plant types without requiring any new data collection or annotation.

Keywords: Open-Set Object Detection · Anomaly Detection · Class-Holdout Evaluation · Weed Detection · Precision Agriculture



Problem Statement

CropAndWeed contains 8,034 images and 111,953 labelled plant instances from 74 species, with a long-tailed frequency distribution: a subset of crops and common weeds account for the large majority of instances, while dozens of rarer species (in some cases fewer than 20 labelled instances) make up a long tail. This project treats the frequent species as the closed set that a detector is trained on, and the rare species as a stand-in for unknown plants a robot might encounter in the field. The task is to build a detector that both performs well on the

known species and reliably flags detections belonging to the held-out rare species as anomalous, without being given any labelled examples of what those rare species actually are.

Proposed Approach

The student will work with the CropAndWeed dataset, first defining a known/unknown species split based on each species' instance count in the existing annotations. Frequent crops and weeds form the closed training set, while the long tail of rare species is held out entirely as the anomaly evaluation set, requiring no new data collection since the split is a direct filter over the dataset's existing per-image annotations. The student will then train a modern object detector on the known-species set and apply and compare anomaly-scoring mechanisms on top of its outputs to flag detections that do not match any known species. Performance will be evaluated both on closed-set detection accuracy for the known species and on open-set separability between known and held-out-rare detections.

References to Get Started

- [1] [The CropAndWeed Dataset: A Multi-Modal Learning Approach for Efficient Crop and Weed Manipulation](#)
- [2] [SegmentMeIfYouCan: A Benchmark for Anomaly Segmentation](#)
- [3] [The Overlooked Elephant of Object Detection: Open Set](#)
- [4] [Expanding Low-Density Latent Regions for Open-Set Object Detection \(OpenDet\)](#)
- [5] [Open-Set Object Detection: Towards Unified Problem Formulation and Benchmarking](#)